

Week 1 Worksheet: Introduction to Quantum Computing - ANSWER KEY

Cornell Quantum Computing Association

Name: _____

Date: _____

Overview Questions

1. What are the two fundamental states that form the computational basis in quantum computing?
 $|0\rangle$ and $|1\rangle$ (also written as $|0\rangle$ and $|1\rangle$)
2. In your own words, explain what makes a quantum coin different from a classical coin.
A classical coin must be in either heads OR tails (definite state), while a quantum coin can be in a superposition - simultaneously both heads AND tails until measured. When measured, the quantum coin collapses to a definite classical outcome.
3. What happens when you measure a qubit that's in superposition?
The superposition collapses and you get a definite classical outcome ($|0\rangle$ or $|1\rangle$) with probabilities determined by the quantum amplitudes.

Classical Foundation

4. If you have 4 classical coins, how many total arrangements are possible?
 $2^4 = 16$ arrangements
5. Write the vector representation for a classical bit in state 1:
 $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$
6. True or False: Classical probability (like "50% heads, 50% tails") represents genuine uncertainty about the physical state of the system.
False. Classical probability represents our ignorance/lack of knowledge. The coin is actually in a definite state; we just don't know which one.

Quantum Concepts

7. Complete the quantum state notation. If a classical bit in state 0 is written as $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, then in quantum notation this is: _____
 $|0\rangle$
8. A qubit is in the state $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$.
 - (a) What is the probability of measuring this qubit in state $|0\rangle$?
 $|\frac{1}{\sqrt{2}}|^2 = \frac{1}{2} = 50\%$

- (b) What is the probability of measuring this qubit in state $|1\rangle$?

$$|\frac{1}{\sqrt{2}}|^2 = \frac{1}{2} = 50\%$$

- (c) Verify that these probabilities sum to 1.

$$\frac{1}{2} + \frac{1}{2} = 1 \quad \checkmark$$

9. Multiple choice: The normalization condition $|\alpha|^2 + |\beta|^2 = 1$ ensures that:

- (a) The quantum state has the right energy
- (b) The measurement probabilities sum to 1
- (c) The qubit rotates properly
- (d) The quantum computer runs faster

10. What is the key mechanism that allows quantum algorithms to provide computational advantage despite measurement collapse?

Quantum interference - wrong answers interfere destructively and cancel out, while correct answers interfere constructively and become more likely to be measured.

Synthesis and Extension

11. Consider a problem where you need to find one specific arrangement out of 2^{20} possible arrangements.

- (a) How many steps would a classical computer need in the worst case?

$$2^{20} = 1,048,576 \text{ steps (might have to check every arrangement)}$$

- (b) How many qubits would you need to represent all possibilities in superposition?

$$20 \text{ qubits (since } 2^{20} \text{ possibilities)}$$

- (c) Why can't you just put the qubits in superposition, measure them, and immediately get the right answer?

Measurement collapses the superposition to just one random outcome. You need to design interference patterns so the correct answer is amplified and wrong answers are suppressed before measurement.

12. Thought experiment: Imagine you have a quantum coin that's in the superposition state $|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$.

- (a) Is this a valid quantum state? Show your work by checking the normalization condition.

$$\text{Check: } |\frac{1}{\sqrt{3}}|^2 + |\sqrt{\frac{2}{3}}|^2 = \frac{1}{3} + \frac{2}{3} = 1 \quad \checkmark \text{ Yes, it's valid.}$$

- (b) If you measured this coin 300 times, approximately how many times would you expect to get outcome 0?

$$300 \times \frac{1}{3} = 100 \text{ times}$$

- (c) How is this different from a classical coin that lands heads with probability $\frac{1}{3}$?

The classical coin is always in a definite state (heads or tails) - the probability just reflects our ignorance. The quantum coin is genuinely in superposition (both states simultaneously) until measured.

Looking Ahead

13. What matrix represents the quantum NOT gate (X gate)?

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

14. If qubits can be represented as vectors, what can quantum gates be represented as?

Matrices (that act on the qubit state vectors)